

Exact Multi-Colour (27-Colour) Parallel Simulated Annealing for the BIEO Energy with a Local Enclosure Term

Abstract

The BIEO enclosure penalty, as implemented (`enclosure_cost`), assigns to every voxel v a bias that fires only when *all* face-neighbours of v belong to $\{\text{inner}, \text{outer}\}$ with at least one outer neighbour present. This is a genuine, non-separable function of v 's entire closed neighbourhood $N[v]$ (a clique of size up to 7 in a 3D 6-connected grid), not a sum of edge terms, and it correctly encodes "no legitimate escape route" rather than mere adjacency — a property ordinary pairwise Markov random fields cannot express. We show that the ordinary two-colour (black/white) checkerboard decomposition is *not* exact for such a term, and we replace it with a 27-colour decomposition of the voxel grid — a proper colouring of the graph square G^2 — under which the same Hammersley–Clifford-type factorization, detailed-balance and annealing arguments go through exactly, at the cost of 27 sequential (but still massively parallel) half-steps per sweep instead of 2. We also correct a second, independent issue in the reference implementation: the update rule must include not only a voxel's own enclosure term but also the induced change in each of its neighbours' enclosure terms, and we give a closed-form, $O(1)$ -per-neighbour incremental formula for this correction.

1 Setup

Let $\Lambda = \{1, \dots, n_1\} \times \{1, \dots, n_2\} \times \{1, \dots, n_3\} \subset \mathbb{Z}^3$ be the voxel grid, $V = \Lambda$, and

$$E = \{\{u, v\} \subset \Lambda : \|u - v\|_1 = 1\}$$

the axis-aligned (6-connected) adjacency, so $d_G(u, v) = \|u - v\|_1$ for the graph distance on $G = (\Lambda, E)$. For $v \in \Lambda$ let $\partial v = \{u : \{u, v\} \in E\}$ (up to 6 face-neighbours; fewer at the volume boundary) and $N[v] = \{v\} \cup \partial v$ its *closed* neighbourhood.

Let L be the label set, $x : \Lambda \rightarrow L$ a configuration, $\Omega = L^\Lambda$. The BIEO objective is

$$\mathcal{E}(x) = U(x) + A(x) + R(x), \quad U(x) = \sum_v c_{v,x_v}^{(1)}, \quad A(x) = \sum_{\{u,v\} \in E} c_{uv,x_u,x_v}^{(2)},$$

with U, A unary/pairwise as before. The enclosure term, exactly as implemented in `enclosure_cost`, is

$$R(x) = \sum_{v \in \Lambda} \sum_{(I,O) \in \mathcal{F}} g_v^{(I,O)}(x_v, x_{\partial v}), \quad (1)$$

where $\mathcal{F}$ is the set of forbidden (inner, outer) label pairs and, writing $\text{esc}(v) = |\{u \in \partial v : x_u \notin \{I, O\}\}|$ (`escape_count`) and $\text{self}(v) = |\{u \in \partial v : x_u = I\}|$ (`self_neighbor_count`),

$$g_v^{(I,O)}(l, x_{\partial v}) = C_{\text{enclose}} \cdot \mathbb{1}[l = I] \cdot \mathbb{1}[\text{esc}(v) = 0] \cdot \mathbb{1}[\text{self}(v) \neq 6] \cdot \mathbb{1}[v \notin \partial \Lambda], \quad (2)$$

$\partial \Lambda$ denoting the set of volume-boundary voxels (`compute_boundary_mask`). Crucially, $g_v^{(I,O)}$ depends only on x_v and $x_{\partial v}$, i.e. its scope is exactly $N[v]$.

Remark 1 (Why R is not — and should not be — pairwise). Fix v and two neighbours $u_1, u_2 \in \partial v$, with the other neighbours held fixed so that $\text{esc}(v) = 0$ depends only on $x_{u_1}, x_{u_2} \in \{O, \star\}$ ($\star$ any label outside $\{I, O\}$). Writing $h(x_{u_1}, x_{u_2}) = \mathbb{K}[\text{esc}(v) = 0]$,

$$h(O, O) = 1, \quad h(O, \star) = h(\star, O) = 0, \quad h(\star, \star) = 0,$$

so the mixed difference $h(O, O) - h(O, \star) - h(\star, O) + h(\star, \star) = 1 \neq 0$: $\text{esc}(v) = 0$ is a genuine conjunction over ∂v and cannot be written as $\sum_{u \in \partial v} f(x_v, x_u)$ for any f . This is not a defect: ”no legitimate escape route anywhere around v ” is definitionally a joint property of the whole neighbourhood. Forcing it into a pairwise form (e.g. ”forbid I and O from touching directly”) would forbid the many legitimate direct I – O contacts that are part of a correct segmentation (e.g. a lumen voxel legitimately touching its own membrane) and is *not* the constraint intended here. We therefore keep R exactly as in (1)–(2) and instead generalize the sampler.

2 A proper colouring for higher-order local terms

Lemma 1 (Clique diameter). *For every $v \in \Lambda$ and $(I, O) \in \mathcal{F}$, the term $g_v^{(I, O)}$ has scope $N[v]$, and $\text{diam}_G(N[v]) \leq 2$ (any two elements of $N[v]$ are at graph distance ≤ 2 , via v).*

Proof. Immediate from $N[v] = \{v\} \cup \partial v$ and the triangle inequality. $\square$

Lemma 2 (27-colouring of Λ). *Define $c : \Lambda \rightarrow \{0, 1, 2\}^3$ by $c(v) = (v_1 \bmod 3, v_2 \bmod 3, v_3 \bmod 3)$, and let $\Lambda_\gamma = c^{-1}(\gamma)$ for $\gamma \in \{0, 1, 2\}^3$ (27 classes, $\Lambda = \bigsqcup_\gamma \Lambda_\gamma$). Then any two distinct u, v with $c(u) = c(v)$ satisfy $d_G(u, v) \geq 3$.*

Proof. $c(u) = c(v)$ means $u - v \in 3\mathbb{Z}^3$, i.e. $u - v = 3m$ for some $m \in \mathbb{Z}^3 \setminus \{0\}$ (as $u \neq v$). Hence $\|u - v\|_1 = 3\|m\|_1 \geq 3$, and $d_G(u, v) = \|u - v\|_1 \geq 3$. $\square$

Corollary 1 (No colour class shares a clique). *For every $v' \in \Lambda$ and every colour class Λ_γ , $|N[v'] \cap \Lambda_\gamma| \leq 1$.*

Proof. If $u, w \in N[v'] \cap \Lambda_\gamma$ with $u \neq w$, then by Lemma 1 $d_G(u, w) \leq 2$, contradicting Lemma 2 ($d_G(u, w) \geq 3$ for same-colour $u \neq w$). $\square$

Remark 2. 27 is a convenient, fully verified upper bound, not a claimed optimum. A denser sublattice of $\mathbb{Z}^3$ with minimum ℓ_1 -norm ≥ 3 (in the spirit of Lee-metric sphere-packing / perfect-code constructions) would give fewer colours; we do not pursue this here and use the simple, easily audited construction above.

3 Generalized local energy and factorization

For $v \in \Lambda$ define the *corrected* local energy

$$\mathcal{E}_{\text{loc}}(l \mid x_{\Lambda \setminus \{v\}}) := \underbrace{c_{v,l}^{(1)} + \sum_{u \in \partial v} c_{uv,l,x_u}^{(2)}}_{U,A \text{ contribution}} + \underbrace{\sum_{(I,O) \in \mathcal{F}} g_v^{(I,O)}(l, x_{\partial v})}_{v\text{'s own enclosure term}} + \underbrace{\sum_{w \in \partial v} \sum_{(I,O) \in \mathcal{F}} g_w^{(I,O)}(x_w, x_{\partial w} \mid v=l)}_{\text{induced change in each neighbour's term}}. \quad (3)$$

The last group is the correction identified as missing from a naively ”self-only” implementation: since $v \in \partial w$ for every $w \in \partial v$, flipping x_v also changes $\text{esc}(w)$ and $\text{self}(w)$ by exactly ± 1 ,

$$\Delta \text{esc}(w) = \mathbb{K}[l \notin \{I, O\}] - \mathbb{K}[x_v \notin \{I, O\}], \quad \Delta \text{self}(w) = \mathbb{K}[l = I] - \mathbb{K}[x_v = I],$$

so this term is computable in $O(|\partial v| \cdot |\mathcal{F}|)$ time from cached $\text{esc}(w), \text{self}(w)$ values, without recomputing the `conv3d` passes.

Proposition 1 (Additivity check). $\mathcal{E}(y) - \mathcal{E}(x) = \mathcal{E}_{\text{loc}}(y_v \mid x_{-v}) - \mathcal{E}_{\text{loc}}(x_v \mid x_{-v})$ for any x, y differing only at v .

Proof. Every term of $\mathcal{E} = U + A + R$ not involving v cancels. Terms involving v are exactly: v 's own unary/pairwise contribution, v 's own $g_v^{(I,O)}$ (scope $N[v] \ni v$), and $g_w^{(I,O)}$ for $w \in \partial v$ (scope $N[w] \ni v$ since $v \in \partial w$) — no other g_c has v in its scope, by Lemma 1 applied at c : $v \in N[c]$ iff $c \in N[v]$, i.e. $c = v$ or $c \in \partial v$. Summing exactly reproduces (3). $\square$

Theorem 1 (Colour-class conditional factorization). Fix $T > 0$, $\pi_T(x) \propto \exp(-\mathcal{E}(x)/T)$. For every colour class Λ_γ ,

$$\pi_T(x_{\Lambda_\gamma} \mid x_{\Lambda \setminus \Lambda_\gamma}) = \prod_{v \in \Lambda_\gamma} \pi_T(x_v \mid x_{\Lambda \setminus \{v\}}), \quad \pi_T(l \mid x_{\Lambda \setminus \{v\}}) \propto \exp(-\mathcal{E}_{\text{loc}}(l \mid x_{\Lambda \setminus \{v\}})/T).$$

Proof. By Corollary 1, every clique $N[c]$ ($c \in \Lambda$) intersects Λ_γ in at most one point. Splitting $\mathcal{E}(x) = \sum_c [\psi_c(x_c) + \sum_{(I,O)} g_c^{(I,O)}(x_c, x_{\partial c})] + \sum_{\{u,v\} \in E} \phi_{uv}(x_u, x_v)$ (unary + all higher-order self-terms + pairwise), and grouping by which, if any, Λ_γ -coordinate each term touches: unary/enclosure terms centred at $c \notin \Lambda_\gamma$ are constants (in x_{Λ_γ}) if $N[c] \cap \Lambda_\gamma = \emptyset$, or depend on exactly the single Λ_γ -coordinate in $N[c] \cap \Lambda_\gamma$ otherwise (by Corollary 1); terms centred at $v \in \Lambda_\gamma$ depend on x_v and on $x_{\partial v} \subset \Lambda \setminus \Lambda_\gamma$ (fixed, since ∂v is at distance 1 from v , hence a different colour by Lemma 2); pairwise terms ϕ_{uv} likewise have exactly one endpoint in Λ_γ whenever the edge touches Λ_γ at all (again by Lemma 2, distance-1 pairs cannot share a colour). Hence, with $x_{\Lambda \setminus \Lambda_\gamma}$ fixed,

$$\mathcal{E}(x) = \text{const}(x_{\Lambda \setminus \Lambda_\gamma}) + \sum_{v \in \Lambda_\gamma} \mathcal{E}_{\text{loc}}(x_v \mid x_{\Lambda \setminus \{v\}}),$$

by Proposition 1 applied coordinatewise. Exponentiating and normalizing over the product space L^{Λ_γ} gives the claim. $\square$

This is the exact generalization, to bounded-degree higher-order local terms and a 27-colouring, of Proposition 1 in *Checkerboard-Decomposition Parallel Simulated Annealing* (which is the special case $\mathcal{F} = \emptyset$, 2 colours).

4 Exactness of one multi-colour sweep

Assumption 1. $q(\cdot \mid l)$ has full support on L and $q_0 := \min_l q(l \mid l) > 0$.

Theorem 2 (Exactness at fixed temperature). Under Assumption 1, let $K_T^{(\gamma)}$ be the half-step kernel for colour class Λ_γ (line 4–6 above) and $K_T = K_T^{(\gamma_{27})} \circ \dots \circ K_T^{(\gamma_1)}$ the full-sweep kernel. Then:

- (i) each $K_T^{(\gamma_i)}$ leaves π_T invariant;
- (ii) K_T leaves π_T invariant;
- (iii) K_T is irreducible and aperiodic on Ω , hence π_T is its unique stationary law and $K_T^k(x, \cdot) \rightarrow \pi_T$.

Algorithm 1 27-Colour Checkerboard-Parallel SA (CPSA-27)

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1: Fix a total order  $\gamma_1, \dots, \gamma_{27}$  of  $\{0, 1, 2\}^3$ .
2: while  $T > T_{\min}$  do
3:   for  $i = 1, \dots, 27$  do
4:     parallel for  $v \in \Lambda_{\gamma_i}$ :
5:       propose  $x'_v \sim q(\cdot \mid x_v)$ ; compute  $\Delta \mathcal{E}_v$  via (3)
6:       accept  $x_v \leftarrow x'_v$  w.p.  $\min\{1, e^{-\Delta \mathcal{E}_v/T}\}$ 
7:     BARRIER()
8:   end for
9:    $T \leftarrow \text{Cool}(T)$ 
10: end while
```

Proof. (i) By Theorem 1, $\pi_T(\cdot \mid x_{\Lambda \setminus \Lambda_\gamma})$ is a product measure over Λ_γ ; $K_T^{(\gamma)}$ applies an independent Metropolis–Hastings step per coordinate targeting exactly this product, hence is reversible w.r.t. it for every fixed $x_{\Lambda \setminus \Lambda_\gamma}$, and (as in the two-colour case) this implies invariance of the full π_T . (ii) Composition of π_T -invariant kernels. (iii) As before: Assumption 1 (full support) gives $K_T(x, y) > 0$ for all x, y (propose/accept every required single-site change in its designated colour-step); finite-state theory gives the rest. $\square$

5 Annealing convergence

Let $\mathcal{N}$ (single-coordinate moves), $\Omega^* = \arg \min \mathcal{E}$, and Hajek’s depth d^* be as before (unchanged: they depend only on $\mathcal{E}$ and $\mathcal{N}$, not on the sweep’s internal structure).

Lemma 3 (Bounded distortion, 27-colour case). *Under Assumption 1, there is $\rho = q_0^{24} \in (0, 1]$, independent of T, k, N , such that for all $T > 0$ and $x \sim_{\mathcal{N}} y$,*

$$\rho \exp\left(-\frac{\mathcal{E}(y) - \mathcal{E}(x)}{T}\right) \leq \frac{K_T(x, y)}{K_T(y, x)} \leq \rho^{-1} \exp\left(-\frac{\mathcal{E}(y) - \mathcal{E}(x)}{T}\right).$$

Proof sketch. Let x, y differ only at $v \in \Lambda_{\gamma_j}$. Colour steps $\gamma_1, \dots, \gamma_{j-1}$: by Corollary 1 and Lemma 1, any site processed here whose local energy could depend on x_v must be within distance 2 of v ; but v has not yet been touched (it is still at its shared, pre-sweep value in both the x - and y -attempt), so these steps are governed by identical probabilities in both directions — no distortion. Step γ_j : sites of Λ_{γ_j} other than v are at distance ≥ 3 from v (Lemma 2), hence, by Lemma 1, their local energy cannot depend on x_v at all; only v ’s own move is affected, and by Proposition 1 and the exact Metropolis–Hastings identity, the ratio of “ $v : x_v \rightarrow y_v$, others stay” to “ $v : y_v \rightarrow x_v$, others stay” equals exactly $\exp(-(\mathcal{E}(y) - \mathcal{E}(x))/T)$. Steps $\gamma_{j+1}, \dots, \gamma_{27}$: only sites c with $d_G(c, v) \leq 2$ can have local energy depending on x_v (Proposition 1’s proof); there are at most $|B_2(v) \setminus \{v\}| = 6 + 18 = 24$ such sites in a 6-connected 3D grid (6 at distance 1, 18 at distance exactly 2: 12 two-axis-diagonal + 6 axis-double-step). Each contributes a “stay” probability ratio (between the post-step- γ_j configurations x and y , which are swapped exactly at v) lying in $[q_0, q_0^{-1}]$ (Assumption 1), by the same argument as the two-colour case. Multiplying the (at most 24) bounded factors gives $\rho = q_0^{24}$. $\square$

Theorem 3 (Annealing convergence, 27-colour sweep). *Under Assumption 1, let $X_{k+1} \sim K_{T_k}(X_k, \cdot)$ with $T_k \downarrow 0$. If $\liminf_k T_k \log k > d^*$, then $\Pr[X_k \notin \Omega^*] \rightarrow 0$. In particular $T_k = \Gamma / \log(k + 2)$, $\Gamma > d^*$, suffices, with the same d^* as classical sequential single-site SA and as the two-colour case*

treated earlier — using 27 colours changes only the constant ρ in Lemma 3, not the asymptotically governing quantity d^* .

Proof. Identical to the two-colour argument (Hajek 1988; extension to bounded multiplicative distortion as in Tsitsiklis 1989), now invoking Lemma 3 in place of the two-colour distortion lemma. $\square$

6 Cost of exactness

One sweep now performs 27 sequential half-steps (barriers) instead of 2. Each half-step still updates $|\Lambda_\gamma| \approx N/27$ voxels in a single massively parallel kernel launch, so the total arithmetic work per sweep is essentially unchanged (still $O(N)$); what grows by a factor $\approx 13.5\times$ is the number of synchronization barriers / kernel launches. On a GPU where each individual launch is cheap relative to the per-voxel work (as is typical for the `conv3d`-based `esc(·)`, `self(·)` computation used here), this is a modest, predictable overhead — not an asymptotic complexity change — in exchange for an exact sampler for the enclosure prior as actually specified (escape-based, not adjacency-based).

1. **Colour order.** Any fixed order of the 27 classes works; we found no requirement to optimize it for correctness.
2. **Incremental reciprocal term.** As noted after (3), the "induced neighbour change" term is a cheap $O(\deg(v))$ correction using cached `esc(w)`, `self(w)` values and does not require extra `conv3d` passes.
3. **Scope of the guarantee.** As before, *Constrained Greedy Dilation* is a separate deterministic post-processing step, outside the scope of this stochastic-optimization argument.
4. **Logarithmic vs. geometric cooling.** The convergence guarantee of Theorem 3 is asymptotic under logarithmic cooling; the finite, geometric schedule used in the experiments ($T_0 = 1 \rightarrow T_{\text{final}} = 10^{-4}$ over 500 sweeps) is the standard practical trade-off (Kirkpatrick et al., 1983) and should be described as such, not as a finite-sample optimality guarantee.

References

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